

Modelling Outbreak Spread Fronts Instead of Per-Pixel Anomalies

BUCKET 10 - CHANGE DETECTION AND EARLY WARNING

For EU plant-health authorities and agronomists (or Malaysian/Indonesian oil-palm estate managers, evidence permitting) who today only learn where pest or disease damage already is from per-parcel satellite anomaly scores, a graph neural network learns a spatial propagation operator over a parcel-adjacency graph built from Sentinel time series so it forecasts which currently-healthy parcels will become newly affected in the next revisit cycle, with success measured by next-cycle newly-affected-parcel prediction rather than by current-state detection accuracy.

VERDICT

VIABLE, with the geography choice being the decisive open question. The EU *Xylella fastidiosa* instantiation passes all hard constraints with a genuinely defensible (if narrow) novelty delta against the closest prior art (Camino et al. 2021) and a confirmed, if not fully resolution-verified, dated outbreak database. The Malaysia oil-palm/rice instantiation explicitly named in the brief currently fails the 'does a dated, spatially-resolved outbreak record exist' check in this search - only aggregate incidence statistics were found - so it should not be assumed viable without first securing a real data partnership (e.g. with MPOB or a plantation operator).

Confidence: Medium. The literature search (arXiv, OpenAlex, Crossref, dblp) consistently surfaced the same closest-prior-art cluster (Camino et al. 2021, Newlands 2018, Hamer et al. 2020, Piou & Marescot 2023) across multiple independent query phrasings, giving reasonable confidence that no closer 2024-2026 GNN-based match was missed. Confidence in the EU data-access route would rise materially with a direct download and inspection of the actual EFSA/Zenodo *Xylella* occurrence records to confirm their spatial/temporal resolution, and confidence in the Malaysia route would only rise via direct contact with MPOB or a Malaysian plantation operator, which was outside the scope of this search pass.

Hard constraint check

Constraint	Verdict	Reasoning
H1 • Novelty	CAVEAT	genuine methodological delta (learned graph propagation operator vs. mechanistic/geostatistical spread models in the cited prior art; next-cycle forecast vs. current-state detection as the evaluation target), but the delta must be argued precisely against Camino et al. (2021), the closest and most threatening prior art found.
H2 • Not generic	PASS	not any of the 8 banned patterns; explicitly frames itself against per-pixel/per-parcel anomaly detection as the thing it is NOT .
H3 • Imagery is primary input	PASS	Sentinel-1/2 imagery-derived per-parcel time series are the node features driving the model; sensor, resolution, revisit, and licence documented above.

H4 • Geography & user	PASS	for the EU Xylella instantiation (named user: EU olive co-ops/plant-health authorities, with a confirmed dated outbreak database); WEAKER/uncertain for the named Malaysia oil-palm/rice instantiation, which lacks a confirmed public ground-truth record - the brief's Malaysia framing should be treated as aspirational pending a real data partnership, not assumed viable.
H5 • Compute & budget	PASS	small graph-based model, small total dataset (15-35GB), all free-tier data, comfortably under the 24GB/12h/RM200/200GB ceilings, modulo the PyTorch Geometric MPS maturity check.
H6 • Pitch quality	PASS	'forecast which parcels get hit next, not just where damage already is' is a clear 60-second pitch, and the map-based next-cycle-prediction-vs-actual-outcome demo is a strong, concrete, live artifact.

The problem

Satellite-based pest/disease monitoring in agriculture almost universally treats each pixel or parcel as an independent classification/anomaly-detection problem: a parcel is scored as healthy or damaged based on its own spectral/SAR time series, with no explicit model of how the causal agent (an insect vector, a windborne fungal spore, an insect population front) propagates across the landscape from already-affected neighbours to still-healthy neighbours. This means existing systems detect damage only after it has already occurred in a given parcel, which is too late for a co-op or plant-health authority with limited spraying/removal resources to act pre-emptively; what they actually need is a forecast of which specific parcels are at elevated risk in the next revisit cycle, so intervention can be targeted ahead of the front rather than in response to it.

Proposed technical approach

Represent the study area as a graph where nodes are agricultural parcels (from a parcel-boundary dataset) and edges encode spatial adjacency/proximity (and optionally wind-direction-weighted or vector-dispersal-distance-weighted connectivity). Each node carries a per-revisit-cycle feature vector derived from Sentinel-1/2 time series (backscatter and vegetation-index anomaly scores, current affected/unaffected state). A spatio-temporal graph neural network (e.g. a graph convolutional or graph attention layer stacked with a temporal recurrent/attention layer, in the style of traffic-forecasting STGNNs) is trained end-to-end to learn a propagation operator: given the graph state (affected/at-risk scores) at cycle t , predict which currently-unaffected neighbouring parcels will newly become affected by cycle $t+1$. Training signal: supervised on a sequence of dated, spatially-resolved outbreak observations (see ground-truth discussion below) using a binary newly-affected/not-affected label per parcel per cycle, with a loss such as weighted binary cross-entropy to handle class imbalance (most parcels stay unaffected each cycle). Output: for each revisit cycle, a ranked list of currently-healthy parcels by predicted risk of becoming newly affected, which is the actionable product for an intervention team.

Novelty

WHAT THE DELTA ACTUALLY IS

The general strategy of coupling a spatial spread/epidemic model with remote sensing is already precedented and must be acknowledged directly: Camino et al. (2021) already couple a mechanistic epidemic-spread model with RS-derived infection probability for *Xylella fastidiosa*; Newlands (2018) and Hamer et al. (2020) already couple satellite/weather monitoring with spatially-explicit transmission models or geostatistical regionalisation for wheat rust and powdery mildew respectively. None of these, however, use a LEARNED graph neural network with an explicit parcel-adjacency graph, trained end-to-end on

Sentinel imagery-derived node features, and evaluated specifically on next-cycle newly-affected-parcel prediction as the headline metric - they all use either classical mechanistic epidemiological models (fixed dispersal kernels/transmission equations) or geostatistical/independent-location ML, not a data-driven propagation operator that learns the spread pattern directly from the graph-structured data the way spatio-temporal GNNs do in traffic/social-network forecasting. The defensible delta is therefore: replacing a hand-specified mechanistic/geostatistical spread model with a learned graph propagation operator, and replacing current-state detection accuracy with next-cycle-forecast accuracy as the evaluation target. This is a real methodological substitution, not 'applied a known GNN to a new crop' - the technical core (a graph neural network trained to forecast which specific unaffected neighbours become affected next, evaluated against that exact target) does not appear in the cited prior art.

PRIOR ART THAT THREATENS IT

Camino et al. (2021, RSE, DOI 10.1016/j.rse.2021.112420) is the closest existing work and must be treated as a serious threat to the novelty claim if not clearly differentiated: it already demonstrates 'couple a spatial spread model with RS-derived infection probability' for Xylella fastidiosa and reports a concrete accuracy improvement (OA 75% to 80%) from doing so. The delta must be argued carefully as 'learned graph propagation operator vs. mechanistic epidemic model' and 'next-cycle forecast vs. current-state classification', not merely 'we also combine spread modelling with RS' - a panel member who knows this paper would otherwise see it as already solved. Piou & Marescot (2023, Current Opinion in Insect Science, DOI 10.1016/j.cois.2023.101024), a locust-forecasting review, explicitly states that statistical approaches are used for presence/outbreak prediction while mechanistic approaches are used for impact prediction, and calls for better integration - this supports (rather than undermines) the claim that a learned, imagery-driven spatial propagation model remains an open gap, not a solved problem. A systematic search of arXiv, OpenAlex, and dblp for '(spatiotemporal) graph neural network' combined with 'plant disease/pest/crop spread' and 'satellite/remote sensing' returned no directly matching 2024-2026 paper, suggesting the specific combination (learned GNN + Sentinel time series + next-cycle newly-affected-parcel metric) is still open, though the adjacent building blocks (spread modelling + RS fusion; GNN traffic/epidemic forecasting in other domains) are each well precededented separately.

GENERIC-PATTERN CHECK

Confirmed **NOT** a banned pattern: not a chatbot/RAG assistant, not a dashboard/monitoring portal, not leaf-photo disease classification, not a plain NDVI viewer, not tabular yield prediction from weather/soil CSVs, not a rerun of an existing crop-type-classification benchmark, not an API-wrapper mobile app. Could the one-sentence summary describe 50 other FYPs? No - 'forecast next-cycle newly-affected parcels via a learned graph propagation operator, not detect current damage' is a specific, non-trivial technical commitment that most 'satellite pest/disease detection' student projects do not attempt, since it requires an explicit adjacency-graph model and a forecasting (not classification) evaluation target.

Sensors and imagery

Sensor	Resolution	Revisit (days)	Bands	Licence
Sentinel-2	10-20 m	5 (nominal, cloud-degraded in practice)	Red/NIR/red-edge vegetation indices for per-parcel anomaly/stress scoring (the node feature)	Copernicus free and open data policy

Sentinel-1	10 m	6-12	VV/VH backscatter, useful where the specific pest/disease affects canopy structure or where persistent cloud cover blocks optical (relevant if a Malaysia/Indonesia site is chosen)	Copernicus free and open data policy
-------------------	------	------	---	--------------------------------------

Imagery-derived per-parcel time series are the node features feeding the graph; the graph structure itself (parcel adjacency) is a static, non-imagery input derived from a parcel-boundary dataset.

Yes - the per-parcel node features that the GNN propagates are derived entirely from Sentinel-1/2 time series (anomaly/stress scores); the graph topology is auxiliary structural information, not a substitute for imagery as the primary sensing input.

Datasets

COPERNICUS DATA SPACE ECOSYSTEM (SENTINEL-1 GRD + SENTINEL-2 L2A)

Access route	dataspace.copernicus.eu via STAC/openEO/Sentinel Hub API
Auth required	Free registration only
Licence	Copernicus free and open data policy, no commercial-use restriction, quotas for bulk use
Approx. size	10-25 for a pilot region/season
Accessible from MY	Yes, verified globally accessible
How verified	WebSearch confirmed dataspace.copernicus.eu live in 2026 with free STAC/openEO/API access

AI4BOUNDARIES (EU PARCEL BOUNDARIES)

Access route	JRC Open Data Catalogue, http://data.europa.eu/89h/0e79ce5d-e4c8-4721-8773-59a4acf2c9c9
Auth required	None
Licence	Open (JRC/EU)
Approx. size	~5-10
Accessible from MY	Yes
How verified	OpenAlex record for d'Andrimont et al. 2023 (essd-15-317-2023) plus WebSearch confirming the live JRC catalogue link

EFSA XYLELLA FASTIDIOSA PEST-OCCURRENCE DATABASE (DATED, GEOREFERENCED OUTBREAK POINTS)

Access route	EFSA Knowledge Junction community on Zenodo (open-access repository); also EFSA's public call-for-data process and Europhyt outbreak notifications
Auth required	Free/open on Zenodo for published database versions; the underlying primary occurrence points were partly collected via an EFSA call-for-data process which may have access restrictions on the rawest layer [uncertain]
Licence	Open access via Zenodo (EFSA Knowledge Junction community)
Approx. size	Small (tabular/point data, not raw imagery, likely well under 1GB)
Accessible from MY	Yes (Zenodo is globally accessible)
How verified	WebSearch confirmed EFSA maintains and updates this database, published in Zenodo, with the most recent (11th) version covering literature through mid-2024 plus Europhyt outbreak notifications; this is the strongest candidate found for a genuine 'dated, spatially-resolved outbreak record'

MPOB / MALAYSIAN OIL-PALM GANODERMA BASAL STEM ROT INCIDENCE RECORDS

Access route	Published survey figures only (e.g. aggregate % incidence and hectares affected by state/estate in journal articles); no public raw, dated, georeferenced parcel-level dataset was located
Auth required	N/A - not accessible in the form needed
Licence	N/A
Approx. size	N/A
Accessible from MY	Aggregate statistics yes, raw georeferenced dated records no
How verified	WebSearch specifically for a public dated/georeferenced MPOB Ganoderma dataset returned only aggregate incidence statistics (e.g. 7.4% BSR incidence, 221,000 ha affected in Malaysian estates) reported in review papers, not a downloadable spatial dataset; this is a genuine negative finding

FAO DESERT LOCUST OBSERVATIONS (DLIS/ELOCUST3, DATA.APPS.FAO.ORG)

Access route	data.apps.fao.org/catalog/dataset/desert-locusts-observations
Auth required	Open catalogue
Licence	Open (FAO)
Approx. size	Small (point observations)
Accessible from MY	Yes
How verified	WebSearch confirmed this FAO catalog dataset exists and is dated/georeferenced from real-time eLocust3/eLocust4 field transmissions; however desert locust (<i>Schistocerca gregaria</i>) outbreaks are concentrated in Africa/West Asia, not Malaysia/Singapore/EU, so this is a proof-of-concept data source rather than a direct match to the H4 user segment

TOTAL DATA VOLUME

Approximately 15-35 GB for a minimum viable pilot combining Sentinel-1/2 time series over a modest AOI, parcel-boundary geometries, and a small tabular outbreak-occurrence dataset - comfortably under the ~200GB practical H5 ceiling.

LABELS AND GROUND TRUTH

This is the single hardest dependency, exactly as the brief anticipates. For the Malaysia/Indonesia oil-palm or rice framing, a systematic search (WebSearch, OpenAlex) did **NOT** find any public, dated, spatially-resolved (parcel- or plot-level) outbreak record for Ganoderma basal stem rot or rice pests - only aggregate regional incidence percentages published in journal reviews, which cannot supply the per-parcel, per-cycle newly-affected labels the GNN needs to train against. This is a genuine negative result: the Malaysia-specific instantiation of this idea is **NOT** currently trainable against real ground truth without a bespoke field-data-collection partnership (e.g. with MPOB, a plantation company, or a local university), which is a significant logistics risk within an FYP timeline. The EU alternative is materially stronger: EFSA maintains a dated, georeferenced *Xylella fastidiosa* pest-occurrence database (via the EFSA Knowledge Junction community on Zenodo, incorporating Europhyt outbreak notifications), which is the closest thing found to the required 'dated, spatially-resolved outbreak record.' A locust alternative also exists via FAO's DLIS/eLocust3 observations but is geographically mismatched to the SEA/EU user requirement (H4). Recommendation: pivot the primary instantiation to *Xylella fastidiosa* spread in EU olive groves (consistent with N2's 'EU is data-rich' steering note) and treat oil-palm/rice-in-Malaysia as an aspirational secondary framing pending a real data-access conversation with MPOB or a Malaysian plantation partner. **[uncertain: whether the EFSA/Zenodo database's spatial/temporal resolution is fine-grained enough for parcel-level next-cycle forecasting, as opposed to broader regional occurrence records - the raw resolution of individual records was not independently verified via direct data download in this pass]**

Baselines to beat

Paper	Reported metric	Dataset
Detection of <i>Xylella fastidiosa</i> in almond orchards by synergic use of an epidemic spread model and remotely sensed plant traits Remote Sensing of Environment 2021 · 10.1016/j.rse.2021.112420	Overall accuracy 80%, kappa 0.48 when coupling a mechanistic spatial epidemic-spread model with an RS-driven SVM infection probability, versus OA 75%/kappa 0.50 for the RS-only model	1,426 almond trees, Spain, airborne hyperspectral/thermal + field campaign
Model-Based Forecasting of Agricultural Crop Disease Risk at the Regional Scale, Integrating Airborne Inoculum, Environmental, and Satellite-Based Monitoring Data Frontiers in Environmental Science 2018 · 10.3389/fenvs.2018.00063	Compares a site-specific stochastic model against a more complex spatially-explicit transmission model for wheat stripe rust; the spatially-explicit model with weather-station-network data had the highest forecast accuracy	Wheat stripe rust (<i>Puccinia striiformis</i>), southern Alberta, Canada

Spatio-Temporal Prediction of the Epidemic Spread of Dangerous Pathogens Using Machine Learning Methods ISPRS International Journal of Geo-Information 2020 · 10.3390/ijgi9010044	68% overall accuracy predicting powdery mildew (<i>Blumeria graminis</i>) infestation events at a daily timescale across a German federal state, with 91% of observed events predicted, using geostatistical regionalisation plus ML, driven by weather/climate data rather than a learned graph propagation operator on imagery	Wind-dispersed fungal pathogen, northern Germany, regional scale
Dynamic Forecast of Desert Locust Presence Using Machine Learning with a Multivariate Time Lag Sliding Window Technique Remote Sensing (MDPI) 2022 · 10.3390/rs14030747	16-day-ahead locust hopper-band presence forecast with 77.46% overall accuracy, ROC-AUC 0.767, using an SVM with time-lagged multi-source satellite/climate features per location, but treating locations independently rather than via a spatial adjacency graph	Desert locust, Somalia/Ethiopia/Kenya, 2000-2020 ground survey + satellite

Evaluation protocol

Primary metric: next-cycle newly-affected-parcel prediction, evaluated as precision/recall/F1 (and ranking metrics such as precision@k or AUPRC, given the expected severe class imbalance since most parcels remain unaffected each cycle) over the set of currently-healthy parcels, comparing the GNN's ranked risk list against which parcels are actually newly reported affected in the following outbreak-record update. This is explicitly **NOT** current-state detection accuracy (which the per-pixel/per-parcel anomaly baselines already report) - the metric only rewards correctly anticipating which specific still-healthy neighbours become affected next, not re-detecting parcels already known to be affected. Baselines for comparison: (a) a naive 'nearest-affected-neighbour' distance-only heuristic (no learning), (b) the per-parcel-independent anomaly-score baseline (no graph propagation, representing current practice), (c) the mechanistic spatial epidemic-spread model of Camino et al. (2021) reimplemented as a non-learned reference. Geographic/temporal generalisation: train the propagation operator on one outbreak season/region, test next-cycle prediction on a held-out season or a spatially disjoint region of the same outbreak.

Feasibility

COMPUTE

Peak memory	Estimated a few GB - a spatio-temporal GNN over a parcel-adjacency graph (typically hundreds to a few thousand nodes for a pilot region) with per-node Sentinel-derived feature vectors is small relative to image-based deep learning models; comparable published STGNN architectures (e.g. traffic forecasting graphs of similar node counts) train well under 24GB
Est. training hours (MPS)	Estimated 2-6 hours on Apple M4 Pro MPS for a pilot-region graph and one outbreak season of Sentinel-derived node features
Fits 24 GB	Yes, comfortably - the graph itself is small; the main memory cost is the upstream Sentinel-1/2 time-series feature extraction, which can be precomputed/cached per parcel rather than held in memory during GNN training
MPS-compatible ops	Standard graph convolution/attention layers (e.g. via PyTorch Geometric) and recurrent/temporal-attention layers are supported on PyTorch MPS, though PyTorch Geometric's MPS support should be spot-checked early as it is less mature than CUDA support for some scatter/gather ops [uncertain]

Colab/Kaggle fallback	Available as a fallback specifically if PyTorch Geometric MPS compatibility issues arise, which is a real early risk to check in week 1
Notes	Estimates are extrapolated from comparable published spatio-temporal GNN architectures in traffic/epidemic forecasting (which use similarly small node counts); not independently benchmarked in this research pass
Cash cost	Approximately RM 0 for core data and compute (Sentinel-1/2 and AI4Boundaries are free; the EFSA/Zenodo outbreak database is free). A modest buffer (~RM 30-50) for Colab/Kaggle GPU credits as a fallback if PyTorch Geometric's MPS support proves immature for the required graph operations. No commercial imagery dependency.
Build : research	Approximately 70:30 engineering:research. The bulk of the effort is building the graph-construction pipeline (parcel geometry ingestion, adjacency/proximity edge definition, optional wind/vector-dispersal weighting), the Sentinel-1/2 per-parcel feature extraction pipeline, and the GNN training/inference/visualisation system; the research component is validating that the learned propagation operator meaningfully outperforms the naive-distance and mechanistic-model baselines on the next-cycle forecasting metric.
Publication venue fit	IGARSS or ICLR ML4RS workshop (strong fit for a spatio-temporal graph learning method applied to Earth observation with a forecasting-not-detection framing); Remote Sensing (MDPI) or Computers and Electronics in Agriculture as a mid-tier journal target if the EU Xylella case study is developed into a full applied paper.

DEMO ARTIFACT

A live laptop demo showing a map of a pilot region's parcel adjacency graph at cycle t , with currently-affected parcels highlighted; the model then produces a ranked heat-map of predicted newly-at-risk neighbouring parcels for cycle $t+1$, and the panel is shown the actual next-cycle outcome overlaid to visually confirm which of the predicted-at-risk parcels indeed became affected - directly demonstrating the 'forecast the front, not detect the state' claim in under two minutes.

User and pain

USER SEGMENT

Primary (best-evidenced): EU plant-health authorities and olive-growing co-ops/agronomists in regions affected by *Xylella fastidiosa* (e.g. Apulia, southern Italy, or other EU olive-growing areas), who would use the parcel-level risk ranking to prioritise limited inspection/removal resources ahead of the advancing front. Secondary/aspirational: Malaysian or Indonesian oil-palm smallholder co-ops and estate agronomists managing *Ganoderma* basal stem rot spread, or rice-farming co-ops managing pest spread in Peninsular Malaysia - named explicitly in the brief, but this instantiation currently lacks a confirmed public, dated, spatially-resolved outbreak record (see `labels_and_ground_truth`), making it the weaker-evidenced option pending a real data partnership.

QUANTIFIED PAIN

Serio et al. (2024, Plants MDPI, DOI 10.3390/plants13111433) reports that *Xylella fastidiosa* subsp. *pauca* has killed millions of olive trees in Apulia, southern Italy, over more than 8,000 square kilometres of infected territory (approximately 40% of the region), and notes that of the research published 2013-2023, only 6.3% addressed treatments while the diagnostic focus (57%) dominated - directly evidencing both the scale of the pain and the current imbalance toward after-the-fact detection over forward-looking spread management. For oil palm, MPOB-linked survey figures cited via WebSearch report 7.4% basal-stem-rot incidence across 221,000 ha of surveyed Malaysian estates (and 9.2%/3,450.70 ha for smallholders), though this is an aggregate incidence statistic rather than a spread-rate or economic-loss

figure specific to the spatial-forecasting use case. **[uncertain: no single figure quantifying the intervention-cost savings from earlier spread forecasting specifically (as opposed to general disease incidence) was found in this search pass]**

Key risks

1. Ground-truth/data-access risk (highest, and geography-dependent): for the Malaysia oil-palm/rice framing, no public, dated, spatially-resolved outbreak record was found - only aggregate regional incidence statistics - making that instantiation untrainable without a bespoke data partnership; the EU Xylella/EFSA route is materially stronger but its raw spatial/temporal resolution was not independently verified.
2. Novelty-framing risk: Camino et al. (2021) already couples a spatial spread model with RS-derived infection probability and reports a concrete accuracy gain from doing so; the delta must be argued precisely as 'learned graph propagation vs. mechanistic model' and 'next-cycle forecast vs. current-state classification', or a panel member familiar with this paper will see it as already done.
3. Evaluation risk: with a graph of at most a few thousand parcels and outbreak events that are typically rare/sparse per cycle, the newly-affected-parcel prediction task will have severe class imbalance, making naive accuracy metrics misleading and requiring careful choice of precision@k/AUPRC-style metrics and a genuinely hard non-learned baseline (nearest-affected-neighbour) to demonstrate real added value.
4. Software-maturity risk: PyTorch Geometric's MPS (Apple Silicon) backend support for scatter/gather graph operations is less mature than its CUDA support and should be spot-checked in week 1, not assumed.
5. Vector/dispersal-mechanism risk: for insect-vectored diseases (Xylella, oil-palm pests), spatial adjacency alone may be a weak proxy for actual spread pathways (insect flight range, wind direction, human-mediated planting-material movement); the graph edges may need non-trivial weighting beyond simple parcel adjacency to be informative, which adds modelling complexity beyond a first version.

Kill criteria

1. Week 1: Attempt to directly download and inspect the EFSA/Zenodo Xylella fastidiosa occurrence database (or the FAO desert-locust observations dataset as a fallback proof-of-concept) and confirm it actually contains per-record dates and coordinates at a resolution fine enough to define 'newly affected this cycle' at the parcel or near-parcel scale; if the resolution is too coarse (e.g. only regional/annual), the specific next-cycle metric as framed cannot be evaluated and the idea needs rescoping.
2. Week 1: Spot-check PyTorch Geometric's MPS compatibility with a toy graph convolution/attention layer on the student's own MacBook Pro M4 Pro; if core scatter/gather ops silently fall back to CPU or error, budget for a Colab/Kaggle fallback immediately rather than discovering this mid-project.
3. Week 2: Reimplement or approximate the Camino et al. (2021) mechanistic spread-model baseline (or a simpler nearest-affected-neighbour heuristic) on the same graph and confirm the intended learned GNN can be set up to be directly compared against it on the next-cycle metric; if a working baseline cannot be stood up, the 'forecasting the front is non-trivial' claim cannot be demonstrated against anything.

Open / unverified items

- labels_and_ground_truth: no public, dated, spatially-resolved outbreak record was confirmed for the Malaysia oil-palm/rice framing named in the brief - only aggregate incidence statistics; the EU EFSA/Zenodo Xylella

database is the strongest alternative found but its exact spatial/temporal record-level resolution was not independently verified via direct download.

- **user_segment:** final geography choice between EU (*Xylella*, better-evidenced ground truth) and Malaysia/Indonesia (oil palm/rice, named in the brief but weaker-evidenced ground truth) is unresolved pending a real data-access conversation.
- **compute_feasibility:** PyTorch Geometric MPS (Apple Silicon) compatibility for the specific graph operations needed was not empirically tested in this pass and should be spot-checked early.
- **quantified_pain:** no single figure quantifying intervention-cost savings specifically from earlier spread forecasting (as opposed to general disease incidence/tree-loss figures) was found.
- **EFSA database access route:** described as available via Zenodo/EFSA Knowledge Junction, but the underlying primary occurrence-point layer was partly assembled via an EFSA call-for-data process whose full public-reuse terms were not independently re-verified via direct WebFetch.

Key references

Detection of *Xylella fastidiosa* in almond orchards by synergic use of an epidemic spread model and remotely sensed plant traits

Remote Sensing of Environment · 2021 · [10.1016/j.rse.2021.112420](https://doi.org/10.1016/j.rse.2021.112420)

The single closest existing prior art; couples a spatial spread model with RS-derived infection probability and reports a concrete accuracy gain - the novelty delta must be argued precisely against this paper.

Model-Based Forecasting of Agricultural Crop Disease Risk at the Regional Scale, Integrating Airborne Inoculum, Environmental, and Satellite-Based Monitoring Data

Frontiers in Environmental Science · 2018 · [10.3389/fenvs.2018.00063](https://doi.org/10.3389/fenvs.2018.00063)

Shows spatially-explicit transmission models already outperform site-specific models for wheat rust forecasting using satellite-based monitoring data, another close precedent for the spread-modelling-plus-RS strategy.

Spatio-Temporal Prediction of the Epidemic Spread of Dangerous Pathogens Using Machine Learning Methods

ISPRS International Journal of Geo-Information · 2020 · [10.3390/ijgi9010044](https://doi.org/10.3390/ijgi9010044)

Demonstrates geostatistical-regionalisation-plus-ML spatial prediction of a wind-dispersed crop pathogen at regional scale, without a learned graph propagation operator - a useful non-GNN baseline.

Spatiotemporal risk forecasting to improve locust management

Current Opinion in Insect Science · 2023 · [10.1016/j.cois.2023.101024](https://doi.org/10.1016/j.cois.2023.101024)

Explicitly identifies the gap between statistical presence-forecasting and mechanistic impact-forecasting approaches, supporting the claim that a learned, imagery-driven spatial propagation model remains an open research direction.

Dynamic Forecast of Desert Locust Presence Using Machine Learning with a Multivariate Time Lag Sliding Window Technique

Remote Sensing (MDPI) · 2022 · [10.3390/rs14030747](https://doi.org/10.3390/rs14030747)

A concrete named baseline for pest-presence forecasting that treats locations independently (no adjacency graph), useful as a non-graph ML comparison point.

A Decade after the Outbreak of *Xylella fastidiosa* subsp. *pauca* in Apulia (Southern Italy): Methodical Literature Analysis of Research Strategies

Plants (MDPI) · 2024 · [10.3390/plants13111433](https://doi.org/10.3390/plants13111433)

Source of the quantified pain figure (millions of olive trees killed, >8,000 sq km infected) and evidence that research has disproportionately focused on diagnosis over forward-looking spread management.

AI4Boundaries: an open AI-ready dataset to map field boundaries with Sentinel-2 and aerial photography

Earth System Science Data · 2023 · [10.5194/essd-15-317-2023](https://doi.org/10.5194/essd-15-317-2023)

Verified free source of EU parcel-boundary geometries needed to construct the adjacency graph.

Generated from the source proposal JSON. Amber [uncertain] flags are carried over verbatim from the source and mark claims that were not independently verified.